How-To Library

Informatica

How to Create Cloud Platform Clusters Using a Workflow in Big Data Management

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Abstract

You can use a workflow to create Hadoop clusters on supported cloud platforms. To implement the cluster workflow, create a Hadoop connection and a cloud provisioning configuration to provide the workflow with the information to connect to the cloud platform and create resources. Then create a workflow with a Create Cluster task, Mapping and other tasks, and a Delete Cluster task. When you include a Delete Cluster task, which terminates and deletes a cloud platform cluster when workflow tasks are complete, the cluster is called an ephemeral cluster.

Supported Versions

- Informatica Big Data Management 10.2.1

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Overview

You can use a workflow to create a cluster that runs Mapping and other tasks on a cloud platform cluster.

A cluster workflow contains a Create Cluster task that you configure with information about the cluster to create. The cluster workflow uses other elements that enable communication between the Data Integration Service and the cloud platform, such as a cloud provisioning configuration and a Hadoop connection.

You can include a Delete Cluster task as the final task in the workflow to terminate the cluster when all workflow tasks are complete. When you include a Delete Cluster task, the cluster is called an ephemeral cluster. The purpose of an

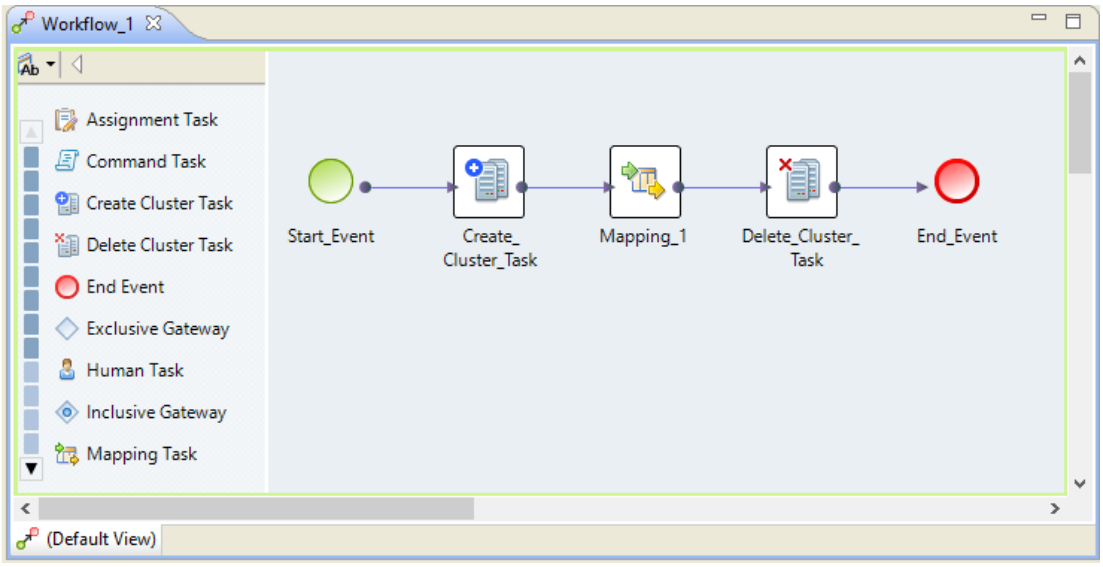
ephemeral cluster is to run cloud cluster resources only when you need them to run jobs, then immediately terminate them to save costs.

You can use a cluster workflow with the Amazon EMR or Microsoft Azure HDInsight cloud platforms.

Cluster Workflow Components

The cluster workflow creates an ephemeral cluster that includes a Create Cluster task, at least one Mapping task, and a Delete Cluster task.

The following image shows a sample cluster workflow:



Create Cluster Task

The Create Cluster task contains all the settings that Amazon EMR or Azure HDInsight requires to create a cluster with a master node and worker nodes. It also contains a reference to a Hadoop connection.

Create one Create Cluster task for a cluster workflow.

When you create a cluster workflow, you drag a Create Cluster task into the workflow editor, and then configure task properties.

Cloud Provisioning Configuration

The cloud provisioning configuration associates the Create Cluster task with the Hadoop connection associated with the workflow.

Configure the cloud provisioning configuration with information about the cloud platform account. To configure the cloud provisioning configuration, use the Administrator tool.

The Create Cluster task must include a reference to the cloud provisioning configuration.

For more information, see the *Big Data Management Administrator Guide*.

Hadoop Connection

Create a dedicated Hadoop connection to use with the cluster.

The Hadoop connection saves property values for the Data Integration Service to use for cluster workflow operations. When you run a cluster workflow, the Data Integration Service creates temporary Hadoop connections based on these values.

Mapping and Other Workflow Tasks

After the Create Cluster task, add other workflow tasks to run on the cluster.

Add one or more Mapping tasks to the workflow. You can also include Command and other workflow tasks.

Before you add mappings to Mapping tasks, you prepare the mappings. Configure the Hadoop Connection property in each mapping to designate whether to run the mapping on the cluster that the workflow creates, or on another cluster.

Delete Cluster Task

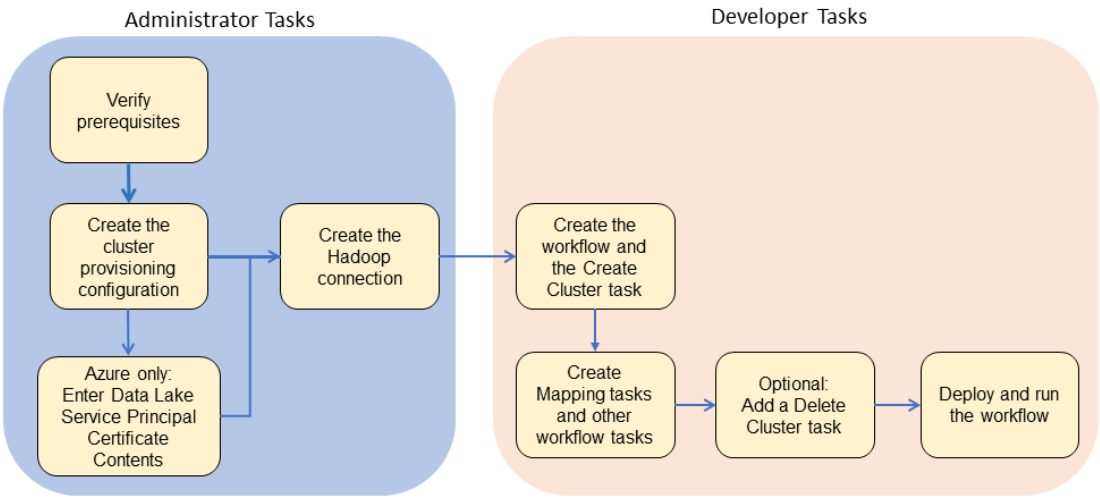
The Delete Cluster task terminates the cluster, and deletes the cluster and other resources that the cluster workflow creates.

To create an ephemeral cluster, add a Delete Cluster task to the workflow. When you do not include it, the cluster continues to run until you terminate it. You can use infacmd ccps deleteCluster to terminate the cluster at any time.

Cluster Workflows Process

Creation of a cluster workflow requires administrator and developer tasks.

The following image shows the process to create, configure, and run a cluster workflow:



First, an administrator completes the following steps:

1. Verify domain and cloud platform prerequisites.
2. Create the cluster provisioning configuration on the domain.
3. Create a Hadoop connection for the cluster workflow to use.

Then, a developer completes the following steps:

1. Create the workflow and the Create Cluster task.
2. Create Mapping tasks and associate them with mappings that you prepared. Optionally add Command tasks and other tasks to the workflow.
3. Optionally add a Delete Cluster task to the workflow.
4. Deploy and run the workflow.

Administrator Tasks

Before the developer can create a cluster workflow, an administrator performs tasks on the Informatica domain.

Before you create a cluster workflow, an administrator must complete the following tasks:

Step 1. Verify prerequisites.

On the cloud platform, install an Informatica domain. Ensure the domain has permission to access cloud storage resources and to create a cluster on the cloud platform. Enable DNS resolution from an on-premises domain.

Step 2. Create the cloud provisioning configuration.

Use the Administrator tool to create a cloud provisioning configuration. The cloud provisioning configuration contains all of the information that the Data Integration Service requires to contact and create resources on the cloud platform.

Step 3. Create a Hadoop connection.

Create a dedicated Hadoop connection to use with the cluster workflow. Populate the Cloud Provisioning Configuration property with the name of the cloud provisioning configuration you created for the workflow. The workflow uses settings in the Hadoop connection to run jobs on the cluster.

Step 1. Verify Prerequisites

Verify cloud platform and domain prerequisites.

Domain Prerequisites

Verify the following prerequisites for the Informatica domain:

- An Informatica domain must be installed. The domain can reside on an instance on the Amazon or Microsoft Azure cloud platform, or on an on-premises machine. If the Informatica instance is installed on-premises, you must configure the VPN to connect to the AWS VPC or Azure vnet where the cluster runs.
- You must have permissions to create connections on the domain.
- To create clusters on AWS, the AWS administrator must open the required ports for the security group to use in the VPC where you want to create the cluster.

Cloud Platform Prerequisites

Verify the following prerequisites for the cloud platform:

- Create a user account with administrator permissions on the cloud platform.
- Create a resource on the cloud platform where you can create clusters.
  - On AWS, create a Virtual Private Cloud (VPC)
  - On Azure, create a Virtual Network (vnet)

  The Informatica domain and the cluster that the workflow creates must be located on this resource.

- If the Informatica domain is installed on-premises, enable DNS resolution.

Enable DNS Resolution from an On-Premises Informatica Domain

If the Informatica domain is installed on-premises, you must enable DNS resolution over the VPN that connects the domain and the cloud platform.

You can enable DNS resolution on Amazon AWS or Microsoft Azure.

Amazon AWS

To run mappings from an on-premises deployment of Big Data Management to a cluster on AWS, you must install and configure Unbound on an EC2 instance. Unbound enables DNS resolution from an on-premises network. To read how to install and configure Unbound in the AWS VPC, see the AWS documentation.

Microsoft Azure

To run mappings from an on-premises deployment of Big Data Management to a cluster on Azure, you must use the Bind utility on the Azure virtural network.

Follow the steps in the Microsoft Azure article "DNS Configuration."

The article gives an example of the contents of the `/etc/bind/named.conf.options` file. You can put a list of available IP addresses on the domain network in the goodclients portion of the file. The following excerpt shows an example:

```
//Add the IP range of the joined network to this list
acl goodclients {
    1.2.3.0/24; # IP address range of the virtual network
    1.2.4.0/24;
    1.2.5.0/24;
    1.2.6.0/24;
    1.2.3.253;
    1.2.3.254;
    localhost;
    localnets;
};
```

Step 2. Create the Cloud Provisioning Configuration

Create the cloud provisioning configuration and configure it with information that the domain needs to access and create resources on the cloud platform.

The properties to configure in the cloud provisioning configuration depend on the cloud platform.

1. From the **Connections** tab, right-click the Domain node and select **New** > **Connection**.

   The **New Connection** dialog box opens.
2. Choose one of the following cloud provisioning configuration types:
   - AWS Cloud Provisioning Configuration. For AWS cloud provisioning properties and values, see "AWS Cloud Provisioning Configuration Properties" on page 7.
   - Azure Cloud Provisioning Configuration. For Azure cloud provisioning properties and values, see "Azure Cloud Provisioning Configuration Properties" on page 8.
3. Enter property values in the configuration wizard, then click **Finish** to create the cloud provisioning configuration.

The cloud provisioning configuration appears in the list of connections in the **Domain Navigator**.

If you want to use ADLS storage with an Azure HDInsight cluster, you must run infacmd ccps updateADLSCertificate to populate the Data Lake Service Principal Certificate Contents property after you create the Azure cloud provisioning configuration.

AWS Cloud Provisioning Configuration Properties

The properties in the AWS cloud provisioning configuration enable the Data Integration Service to contact and create resources on the AWS cloud platform.

General Properties

The following table describes cloud provisioning configuration general properties:

Property	Description
Name	Name of the cloud provisioning configuration.
ID	ID of the cloud provisioning configuration. Default: Same as the cloud provisioning configuration name.
Description.	Optional. Description of the cloud provisioning configuration.
AWS Access Key ID	Optional. ID of the AWS access key, which AWS uses to control REST or HTTP query protocol requests to AWS service APIs. If you do not specify a value, Informatica attempts to follow the Default Credential Provider Chain.
AWS Secret Access Key	Secret component of the AWS access key. Required if you specify the AWS Access Key ID.
Region	Region in which to create the cluster. This must be the region in which the VPC is running. Use AWS region values. For a list of acceptable values, see AWS documentation. **Note:** The region where you want to create the cluster can be different from the region in which the Informatica domain is installed.

Permissions

The following table describes cloud provisioning configuration permissions properties:

Property	Description
EMR Role	Name of the service role for the EMR cluster that you create. The role must have sufficient permissions to create a cluster, access S3 resources, and run jobs on the cluster. When the AWS administrator creates this role, they select the “EMR” role. This contains the default AmazonElasticMapReduceRole policy. You can edit the services in this policy.
EC2 Instance Profile	Name of the EC2 instance profile role that controls permissions on processes that run on the cluster. When the AWS administrator creates this role, they select the “EMR Role for EC2” role. This includes S3 access by default.
Auto Scaling Role	Required if you configure auto-scaling for the EMR cluster. This role is created when the AWS administrator configures auto-scaling on any cluster in the VPC. Default: When you leave this field blank, it is equivalent to setting the Auto Scaling role to “Proceed without role” when the AWS administrator creates a cluster in the AWS console.

EC2 Configuration

The following table describes cloud provisioning configuration EC2 configuration properties:

Property	Description
EC2 Key Pair	EC2 key pair to enable communication with the EMR cluster master node. Optional. This credential enables you to log into the cluster. Configure this property if you intend the cluster to be non-ephemeral.
EC2 Subnet	ID of the subnet on the VPC in which to create the cluster. Use the subnet ID of the EC2 instance where the cluster runs.
Master Security Group	Optional. ID of the security group for the cluster master node. Acts as a virtual firewall to control inbound and outbound traffic to cluster nodes. Security groups are created when the AWS administrator creates and configures a cluster in a VPC. In the AWS console, the property is equivalent to ElasticMapReduce-master. You can use existing security groups, or the AWS administrator might create dedicated security groups for the ephemeral cluster. If you do not specify a value, the cluster applies the default security group for the VPC.
Additional Master Security Groups	Optional. IDs of additional security groups to attach to the cluster master node. Use a comma-separated list of security group IDs.
Core and Task Security Group	Optional. ID of the security group for the cluster core and task nodes. When the AWS administrator creates and configures a cluster In the AWS console, the property is equivalent to the ElasticMapReduce-slave security group If you do not specify a value, the cluster applies the default security group for the VPC.
Additional Core and Task Security Groups	Optional. IDs of additional security groups to attach to cluster core and task nodes. Use a comma-separated list of security group IDs.
Service Access Security Group	EMR managed security group for service access. Required when you provision an EMR cluster in a private subnet.

Azure Cloud Provisioning Configuration Properties

The properties in the Azure cloud provisioning configuration enable the Data Integration Service to contact and create resources on the Azure cloud platform.

Authentication Details

The following table describes authentication properties to configure:

Property	Description
Name	Name of the cloud provisioning configuration.
ID	ID of the cloud provisioning configuration. Default: Same as the cloud provisioning configuration name.
Description	Optional. Description of the cloud provisioning configuration.
Subscription ID	ID of the Azure account to use in the cluster creation process.

Property	Description
Tenant ID	A GUID string associated with the Azure Active Directory.
Client ID	A GUID string that is the same as the Application ID associated with the Service Principal. The Service Principal must be assigned to a role that has permission to create resources in the subscription that you identified in the Subscription ID property.
Client Secret	An octet string that provides a key associated with the client ID.

Storage Account Details

Choose to configure access to one of the following storage types:

- Azure Data Lake Storage (ADLS). See Azure documentation.
- An Azure Storage Account, known as general or blob storage. See Azure documentation.

The following table describes the information you need to configure Azure Data Lake Storage (ADLS) with the HDInsight cluster:

Property	Description
Azure Data Lake Store Name	Name of the ADLS storage to access. The ADLS storage and the cluster to create must reside in the same region.
Data Lake Service Principal Client ID	A credential that enables programmatic access to ADLS storage. Enables the Informatica domain to communicate with ADLS and run commands and mappings on the HDInsight cluster. The service principal is an Azure user that meets the following requirements: - Permissions to access required directories in ADLS storage. - Certificate-based authentication for ADLS storage. - Key-based authentication for ADLS storage.
Data Lake Service Principal Certificate Contents	The Base64 encoded text of the public certificate used with the service principal. Leave this property blank when you create the cloud provisioning configuration. After you save the cloud provisioning configuration, log in to the VM where the Informatica domain is installed and run infacmd ccps updateADLSCertificate to populate this property.
Data Lake Service Principal Certificate Password	Private key for the service principal. This private key must be associated with the service principal certificate.
Data Lake Service Principal Client Secret	An octet string that provides a key associated with the service principal.
Data Lake Service Principal OAUTH Token Endpoint	Endpoint for OAUTH token based authentication.

The following table describes the information you need to configure Azure General Storage, also known as blob storage, with the HDInsight cluster:

Property	Description
Azure Storage Account Name	Name of the storage account to access. Get the value from the Storage Accounts node in the Azure web console. The storage and the cluster to create must reside in the same region.
Azure Storage Account Key	A key to authenticate access to the storage account. To get the value from the Azure web console, select the storage account, then Access Keys. The console displays the account keys.

Cluster Deployment Details

The following table describes the cluster deployment properties that you configure:

Property	Description
Resource Group	Resource group in which to create the cluster. A resource group is a logical set of Azure resources.
Virtual Network Resource Group	Optional. Resource group to which the virtual network belongs. If you do not specify a resource group, the Data Integration Service assumes that the virtual network is a member of the same resource group as the cluster.
Virtual Network	Name of the virtual network or vnet where you want to create the cluster. Specify a vnet that resides in the resource group that you specified in the Virtual Network Resource Group property. The vnet must be in the same region as the region in which to create the cluster.
Subnet Name	Subnet in which to create the cluster. The subnet must be a part of the vnet that you designated in the previous property. Each vnet can have one or more subnets. The Azure administrator can choose an existing subnet or create one for the cluster.

External Hive Metastore Details

You can specify the properties to enable the cluster to connect to a Hive metastore database that is external to the cluster.

If you do not specify an existing external database in this dialog box, the cluster creates its own database on the cluster. This database is terminated when the cluster is terminated.

You can use an external relational database like MySQL or Amazon RDS as the Hive metastore database. The external database must be on the same cloud platform as the cluster to create.

The following table describes the Hive metastore database properties that you configure:

Property	Description
Database Name	Name of the Hive metastore database.
Database Server Name	Server on which the database resides. **Note:** The database server name on the Azure web console commonly includes the suffix `database.windows.net`. For example: `server123xyz.database.windows.net`. You can specify the database server name without the suffix and Informatica will automatically append the suffix. For example, you can specify `server123xyz`.

Property	Description
Database User Name	User name of the account for the domain to use to access the database.
Database Password	Password for the user account.

Complete the Azure Cloud Provisioning Configuration

When you want to access Azure Data Lake Storage (ADLS) with the cluster workflow, complete the following steps after you configure and save the cloud provisioning configuration for Azure:

1. Log in to the VM where the Informatica domain is installed, and open a command shell.
2. From the command line, issue the following command:

```
/infacmd.sh ccps updateADLSCertificate -dn <domain name> -un <user name> -pd <password> -
cpcid <cloud provisioning connection name>
-certPath <domain location of certificate>
```

   The command automatically populates the Data Lake Service Principal Certificate Contents property of the cloud provisioning connection.

Step 3. Create a Hadoop Connection

Create a dedicated Hadoop connection to use with the cluster workflow.

The Hadoop connection saves property values for the Data Integration Service to use for cluster workflow operations. When you run a cluster workflow, the Data Integration Service uses settings in the Hadoop connection to run jobs on the cluster.

When you configure a Hadoop connection for the cluster workflow, populate the Cloud Provisioning Configuration property with the name of the cloud provisioning configuration you created for the workflow. Leave the Cluster Configuration property blank.

When you create the workflow, populate the Connection Name property of the Create Cluster task with this Hadoop connection.

Creating a Connection

In the Administrator tool, you can create relational database, social media, and file systems connections.

1. In the Administrator tool, click the **Manage** tab.
2. Click the **Connections** view.
3. In the Navigator, select the domain.
4. In the Navigator, click **Actions** > **New** > **Connection**.

   The **New Connection** dialog box appears.
5. In the **New Connection** dialog box, select the connection type, and then click **OK**.

   The **New Connection** wizard appears.
6. Enter the connection properties.

   The connection properties that you enter depend on the connection type. Click **Next** to go to the next page of the **New Connection** wizard.

7. When you finish entering connection properties, you can click **Test Connection** to test the connection.
8. Click **Finish**.

Developer Tasks

After you complete administrator tasks, complete the following tasks in the Developer tool:

Step 1. Create the cluster workflow.

Create the cluster workflow and add and configure workflow elements.

Step 2. Add Mapping and other tasks to the workflow.

Populate the cluster workflow with at least one Mapping task. You can add Command or other workflow tasks and events. Set mapping and Mapping task properties to specify where the workflow runs Mapping tasks.

Step 3. Add a Delete Cluster task.

Add a Delete Cluster task to the workflow to save cloud platform resources when workflow tasks are complete. The Delete Cluster task terminates and deletes the cluster and the Hadoop connections that connect to it.

A cluster created by a workflow and terminated with a Delete Cluster task is called an ephemeral cluster.

Step 4. Deploy and run the workflow.

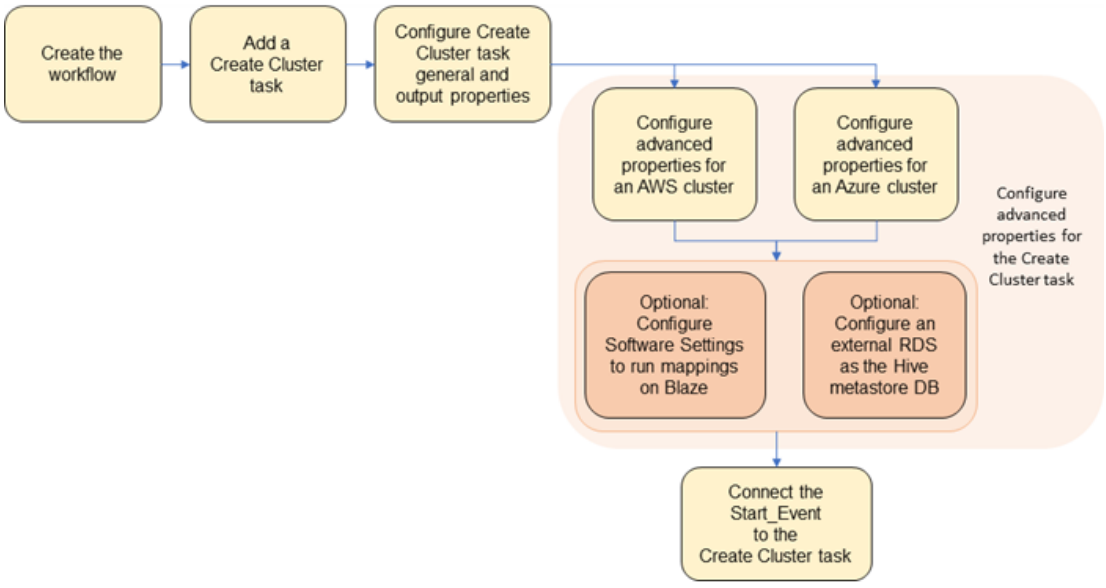
After you complete development of the cluster workflow, deploy and run the workflow.

Step 1. Create the Cluster Workflow and Configure the Create Cluster Task

Create the cluster workflow and add and configure workflow elements.

The cluster workflow must have a Create Cluster task and at least one Mapping task. You can add Command and other workflow tasks. You can also add a Delete Cluster task, so that the cluster becomes an ephemeral cluster.

The following image shows the steps to create the workflow and to create and configure the Create Cluster task:



Configure the Cluster Workflow

A cluster workflow must have one Create Cluster task.

1. In the Developer tool, create a workflow.
2. From the palette of tasks, drag a Create Cluster task to the workflow editor.
3. Complete the Create Cluster task general properties.

Property	Description
Name	Task name.
Description	Optional description.
Connection Name	Name of the cloud provisioning configuration to use with the workflow.
Connection Type	Choose from the following options: - Amazon EMR. Create an Amazon EMR cluster. - HDInsight. Create an Azure HDInsight cluster.

4. Configure task input and output properties.

   Input properties

   The Create Cluster task does not require any unique values for task input properties.

   Output properties

   Set the Cluster Identifier property to the default value, AutoDeployCluster.

   Note: The Cluster Identifier property of the Create Cluster task overrides the Cluster Identifier property of the Mapping task.
5. Set the advanced properties that correspond to your cloud platform.
   - To create a cluster on Amazon Web Services, see "Amazon EMR Advanced Properties for the Create Cluster Task " on page 14.
   - To create a cluster on Microsoft Azure, see "Azure HDInsight Advanced Properties for the Create Cluster Task" on page 16.
6. Configure the Software Settings property in the advanced properties if you want to perform the following optional tasks:
   - To run mappings on the Blaze engine, see "Configure the Create Cluster Task to Run Mappings on the Blaze Engine" on page 17.
   - To configure an external relational database as the Hive metastore database, see "Configure the Cluster to Use an External RDS as the Hive Metastore Database" on page 18.
7. Connect the workflow Start_Event to the Create Cluster task.

Amazon EMR Advanced Properties for the Create Cluster Task

General Options

The following table describes general options that you can set for an EMR cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Release Version	EMR version to run on the cluster. Enter the AWS version tag string to designate the version. For example: `emr-5.8.0` Default is Latest version supported.
Connection Name	Name of the Hadoop connection that you configured for use with the cluster workflow.
S3 Log URI	Optional. S3 location of logs for cluster creation. Format: `s3://<bucket name>/<folder name>` If you do not supply a location, no cluster logs will be stored.

Master Instance Group Options

The following table describes master instance group options that you can set for an EMR cluster:

Property	Description
Master Instance Type	Master node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Master Instance Maximum Spot Price	Maximum spot price for the master node. Setting this property changes the purchasing option of the master instance group to Spot instead of On-demand.

Core Instance Group Options

The following table describes core instance group options that you can set for an EMR cluster:

Property	Description
Core Instance Type	Core node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Core Instance Count	Number of core EC2 instances to create in the cluster. Default is 2.
Core Instance Maximum Spot Price	Maximum spot price for core nodes. Setting this property changes the purchasing option of the core instance group to Spot instead of On-demand.
Core Auto-Scaling Policy	Optional. Auto-scaling policy for core instances. Type the policy JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`

Task Instance Group Options

The following table describes task instance group options that you can set for an EMR cluster:

Property	Description
Task Instance Type	Task node EC2 instance type. You can specify any available EC2 instance type. Default is m4.4xlarge.
Task Instance Count	Number of task EC2 instances to create in the cluster. Default is 2.
Task Instance Maximum Spot Price	Maximum spot price for task nodes. Setting this property changes the purchasing option of the task instance group to Spot instead of On-demand.
Task Auto-Scaling Policy	Optional. Auto-scaling policy for task instances. Type the policy JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`

Additional Options

The following table describes additional options that you can set for an EMR cluster:

Property	Description
Applications	Optional. Applications to add to the default applications that AWS installs. AWS installs certain applications when it creates an EMR cluster. In addition, you can specify additional applications. Select additional applications from the drop-down list. This field is equivalent to the Software Configuration list in the AWS EMR cluster creation wizard.
Tags	Optional. Tags to propagate to cluster EC2 instances. Tags assist in identifying EC2 instances. Format: `TagName1=TagValue1,TagName2=TagValue2`
Software Settings	Optional. Custom configurations to apply to the applications installed on the cluster. This field is equivalent to the Edit Software Settings field in the AWS cluster creation wizard. You can use this as a method to modify the software configuration on the cluster. Type the configuration JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_custom_config_file>`
Steps	Optional. Commands to run after cluster creation. For example, you can use this to run Linux commands or HDFS or Hive Hadoop commands. This field is equivalent to the Add Steps field in the AWS cluster creation wizard. Type the command statement here, or or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_command_file>`

Property	Description
Bootstrap Actions	Optional. Actions to perform after EC2 instances are running, and before applications are installed. Type the JSON statement here, or provide a path to a file that contains a JSON statement. Format: `file:\\<path_to_policy_config_file>`
Task Recovery Strategy	Choose from the following options: - Restart task - Skip task Default is Restart task.

Azure HDInsight Advanced Properties for the Create Cluster Task

The following table describes the Advanced properties for a Microsoft Azure HDInsight cluster:

Property	Description
Cluster Name	Name of the cluster to create.
Azure Cluster Type	Type of the cluster to be created. Choose one of the options in the drop-down list. Default is Hadoop.
HDInsight version	HDInsight version to run on the cluster. Enter the HDInsight version tag string to designate the version. Default is the latest version supported.
Azure Cluster Location	Use the drop-down list to choose the location in which to create the cluster.
Head Node VM Size	Size of the head node instance to create. Default is Standard_D12_v2.
Number of Worker Node Instances	Number of worker node instances to create in the cluster. Default is 2.
Worker Node VM Size	Size of the worker node instance to create. Default is Standard_D13_v2.
Default Storage Type	Primary storage type to be used for the cluster. Choose one of the following options: - Azure Data Lake Store - Azure BLOB storage account Default is BLOB storage
Default Storage Container or Root Mount Path	Default container for data. Type one of the following paths: - For ADLS storage, type the path to the storage. For example, you can type `storage-name` or `storage-name/folder-name`. - For blob storage, type the path to the container. Format: `/path/`

Property	Description
Log Location	Optional. Path to the directory to store workflow event logs. Default is `/app-logs`.
Attach External Hive Metastore	If you select this option, the workflow attaches an external Hive metastore to the cluster if you configured an external Hive metastore in the cloud provisioning configuration.

Bootstrap JSON String	JSON statement to run during cluster creation. You can use this statement to configure cluster details. For example, you could designate a Hadoop connection for the cluster, add tags to cluster resources, or run script actions.
	Choose one of the following methods to populate the property:
	- Type the JSON statement. Use the following format:
	`{ "core-site" : { "<sample_property_key1>": "<sample_property_val1>", "<sample_property_key2>": "<sample_property_val2>" }, "tags": { "<tag_key>": "<tag_val>" }, "scriptActions": [ { "name": "setenvironmentvariable", "uri": "scriptActionUri", "parameters": "headnode" } ] }`
	- Provide a path to a file that contains a JSON statement. Format:
	`file://<path_to_bootstrap_file>`

Configure the Create Cluster Task to Run Mappings on the Blaze Engine

If you want to use the Blaze engine to run mappings on the cloud platform cluster, you must set cluster configuration properties in the Software Setting property of the Create Cluster task.

Configure the Create Cluster task to set configuration properties in *-site.xml files on the cluster. Hadoop clusters run based on these settings.

The following text shows sample configuration of the Software Settings property:

```
[
  {
     "Classification":"yarn-site",
     "Properties":{
     "yarn.scheduler.minimum-allocation-mb":"250",
     "yarn.scheduler.maximum-allocation-mb":"8192",
     "yarn.nodemanager.resource.memory-mb":"16000",
     "yarn.nodemanager.resource.cpu-vcores":"12"
   }
  },
  {
    "Classification":"core-site",
    "Properties":{
    "hadoop.proxyuser.<DIS/OSPUSER>.groups":"<group names>",
    "hadoop.proxyuser.<DIS/OSPUSER>.hosts":"*"
  }
 }
]
```

yarn-site

yarn.scheduler.minimum-allocation-mb

The minimum RAM available for each container. Required for Blaze engine resource allocation.

yarn.scheduler.maximum-allocation-mb

The maximum RAM available for each container. Required for Blaze engine resource allocation.

yarn.nodemanager.resource.memory-mb

The maximum RAM available for each container. Set the maximum memory on the cluster to increase resource memory available to the Blaze engine.

yarn.nodemanager.resource.cpu-vcores

The number of virtual cores for each container. Required for Blaze engine resource allocation.

core-site

hadoop.proxyuser.<proxy user>.groups

Defines the groups that the proxy user account can impersonate. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set to group names of impersonation users separated by commas. If less security is preferred, use the wildcard " * " to allow impersonation from any group.

hadlop.proxyuser.<user>.hosts

Defines the host machines that a user account can impersonate. On a secure cluster the <proxy user> is the Service Principal Name that corresponds to the cluster keytab file. On a non-secure cluster, the <proxy user> is the system user that runs the Informatica daemon.

Set the property to " * " to allow impersonation from any host. This is required to run a Blaze mapping on a cloud platform cluster.

Configure the Cluster to Use an External RDS as the Hive Metastore Database

If you want to use a relational database on the cloud platform as the Hive metastore database for the cluster, you must set cluster configuration properties in the Software Setting property of the Create Cluster task.

Configure the Create Cluster task to set configuration properties in the hive-site.xml configuration file on the cluster. Use a text file to specify hive-site settings, and specify the path to the file in the Software Settings property.

The following text shows sample configuration of the Software Settings property:

```
{
      "Classification":"hive-site",
      "Properties":{
         "javax.jdo.option.ConnectionURL":"jdbc:mysql:\/\/<RDS_HOST>:<PORT>\/<USER_SCHEMA>?
createDatabaseIfNotExist=true",
         "javax.jdo.option.ConnectionDriverName":"<JDBC driver name>",
         "javax.jdo.option.ConnectionUserName":"<USER>",
         "javax.jdo.option.ConnectionPassword":"<USER>"
      }
   }
```

Example:

```
Example:
{
      "Classification":"hive-site",
      "Properties":{
         "javax.jdo.option.ConnectionURL":"jdbc:mysql:\/\/<host name>:<port number>\/hive?
```

```
createDatabaseIfNotExist=true",
         "javax.jdo.option.ConnectionDriverName":"org.mariadb.jdbc.Driver",
         "javax.jdo.option.ConnectionUserName":"hive",
         "javax.jdo.option.ConnectionPassword":"hive"
      }
   }
```

hive-site

javax.jdo.option.ConnectionURL

JDBC connection string for the data store.

javax.jdo.option.ConnectionDriverName

JDBC driver class name for the data store. Specify a JDBC driver that is compatible with the cloud platform.

javax.jdo.option.ConnectionUserName

User name to use to connect to the database.

javax.jdo.option.ConnectionPassword

Password for the database user account.

Step 2. Create Mapping and Other Workflow Tasks

Populate the cluster workflow with at least one Mapping task. You can add Command or other workflow tasks and events.

Before you create a Mapping task, configure mappings to run in the cluster workflow. Then add Mapping tasks to run the mappings that you prepared.

You can also add Command and other tasks to the workflow.

Prepare Mappings for the Cluster Workflow

Prepare mappings for inclusion in the cluster workflow by setting mapping and Mapping task properties to specify where mappings run.

You can create a cluster workflow to run some mappings on the cluster that the workflow creates, and other mappings on another cluster.

Run the mapping on the cluster that the workflow creates.

The following table describes the run-time behavior based on how you configure the mapping and Mapping task:

Mapping **Hadoop Connection Property**	**Mapping Task** **Cluster Identifier Property**	**Run Time Behavior**
Auto Deploy	Auto Deploy	The Data Integration Service generates temporary Hadoop connections based on the values in the Hadoop connection associated with the workflow, and uses the temporary connections to run mappings on the cluster.
<Hadoop connection name>	Auto Deploy	The Mapping task Cluster Identifier property overrides the mapping Hadoop connection property. You might want to do this if you want to maintain the mapping Hadoop connection property value.

Run the mapping on another cluster.

The following table describes the run-time behavior based on how you configure the mapping and Mapping task:

Mapping Hadoop Connection Property	Mapping Task Cluster Identifier Property	Run Time Behavior
<Hadoop connection name>	Blank	The Mapping task Cluster Identifier property takes input from the Hadoop connection and runs the mapping on the cluster identified in the cloud configuration property of the Hadoop connection.

Add Mapping and Other Tasks to the Workflow

1. Drag a Mapping task from the task list to the workflow editor.

   The **Mapping Task** dialog box opens.
2. Name the Mapping task.
3. Select a mapping to run with the Mapping task. Click **Browse** next to the Mapping property, select a mapping, and click **Finish**.
4. Optionally select a parameter set to associate with the Mapping task. Click **Browse** next to the Parameter Set property, select a parameter set, and click **Finish**.

   For more information on how to use parameter sets with mappings, see the *Informatica Developer Mapping Guide*.
5. Optionally complete Input and Output properties.

   The Mapping task does not require any unique values for input or output properties.
6. Configure the Cluster Identifier property in Advanced properties.

   The Cluster Identifier property designates the cluster to use to run the Mapping task.

   The following table describes values for Cluster Identifier properties:

Value	Description
Blank (no value)	Run the mapping runs on the cluster configured in the Hadoop connection associated with the mapping.
AutoDeploy	Run the mapping on the cluster that the workflow creates. When you choose this option, it also populates the Cluster Identifier property in the Create Cluster task with the value Set to AutoDeployCluster. Default is AutoDeploy.
(Assign to task input)	Select this option to accept input from another source than the Create Cluster task. If you choose this option, enter a parameter value in the Cluster Identifier property of the Mapping task Input properties tab.

7. Click **Finish** to the Mapping task.
8. Optionally add more Mapping and other tasks to the workflow.

   You can include any other workflow tasks in a cluster workflow. For example, you might want to add a Command task to perform tasks after a Mapping task runs.

Step 3. Add a Delete Cluster Task

To create an ephemeral cluster, add a Delete Cluster task.

The Delete Cluster task terminates the cluster and deletes the cluster and other resources that the cluster workflow creates.

If you do not add a Delete Cluster task, the cluster that the workflow creates remains running when the workflow ends. You can delete the cluster at any time.

1. Drag a Delete Cluster task to the workflow editor.
2. In the General properties, optionally rename the Delete Cluster task.
3. Connect the final task in the workflow to the Delete Cluster task, and connect the Delete Cluster task to the workflow End_Event.

You can also use infacmd ccps deleteCluster to delete a cloud cluster.

Step 4. Deploy and Run the Workflow

After you complete the cluster workflow, deploy and run the workflow.

You can monitor cluster provisioning tasks on the AWS or Azure web console. If you configured a log location, view the logs at the location that you configured in the Create Cluster task properties.

You can also monitor Data Integration Service jobs in the Administrator tool.

Note: After the workflow begins executing tasks, the task to provision the cluster may take several minutes.

Monitoring Azure HDInsight Cluster Workflow Jobs

You can access mapping log URLs through the **Monitoring** tab in the Administrator tool to monitor workflow jobs that run on an Azure HDInsight cluster. The log location depends on the run-time engine that each mapping uses.

Blaze and Spark engines

To access the monitoring URL for mappings that run on Blaze or Spark, expand the workflow and the mapping in the **Monitoring** tab. Select the Grid Task and view the value for the Monitoring URL property in the lower pane. Use this path to find the log.

Hive engine

To access the monitoring URL for mappings that run on Hive, expand the workflow and the mapping in the **Monitoring** tab. Select a Hive Query job, and then expand the MR Job Details node in the lower pane. The Job ID is hyperlinked, but clicking on the link does not lead to the log. To find the job monitoring log, copy the URL path and read it to find the log. Repeat the steps for each Hive Query job.

infacmd ccps Command Reference

Use infacmd ccps commands to administer cloud platform clusters.

deleteClusters

Deletes clusters created by the cluster workflow from the cloud platform.

The infacmd ccps deleteClusters command uses the following syntax:

```
deleteClusters
<-DomainName|-dn> domain_name
```

```
<-ServiceName|-sn> service_name
<-UserName|-un> user_name
<-Password|-pd> password
<-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
<-ClusterIDs|-cids> cluster_ids
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

Note: When you use this command to delete clusters on the Azure cloud platform, the process blocks any other command through the command shell until the Azure cloud platform completes the process to release cluster resources. This process could take several minutes. If you try to kill the command using CTRL-C, and then re-run the command, the same time delay and block apply.

The following table describes infacmd ccps deleteClusters options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration.
-ClusterIDs -cids	cluster_ids	Required. Comma-separated list of clusters to delete. The cluster ID is the same as the cluster ID listed on the cloud platform site.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.

Option	Argument	Description
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.
-ConfigurationName -cn	Name of the cluster configuration	Required. Name of the cluster configuration on the domain. Values are not case sensitive.
-DeleteConnections -dc	delete_associated_connection	Optional. Deletes the connections that the cluster configuration created. Use one of the following values: - TRUE - FALSE Default is FALSE.

listClusters

Lists clusters that the cluster workflow creates and that exist on the cloud platform.

The infacmd ccps listClusters command uses the following syntax:

```
listClusters
<-DomainName|-dn> domain_name
<-ServiceName|-sn> service_name
<-UserName|-un> user_name
<-Password|-pd> password
<-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
[<-SecurityDomain|-sdn> security_domain]
[<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

The following table describes infacmd ccps listClusters options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.

Option	Argument	Description
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.

updateADLSCertificate

Updates the Azure Data Lake Service Principal certificate path in a cloud provisioning configuration.

The infacmd ccps updateADLSCertificate command uses the following syntax:

```
updateADLSCertificate
    <-DomainName|-dn> domain_name
    <-UserName|-un> user_name
    <-Password|-pd> password
    <-CloudProvisioningConfigurationID|-cpcid> cloud_provisioning_configuration_id
    <-CertificateFilePath|-certPath> certificate_file_path
    [<-SecurityDomain|-sdn> security_domain]
    [<-ResilienceTimeout|-re> timeout_period_in_seconds]
```

The following table describes infacmd ccps updateADLSCertificate options and arguments:

Option	Argument	Description
-DomainName -dn	domain_name	Required. Name of the Informatica domain. You can set the domain name with the -dn option or the environment variable INFA_DEFAULT_DOMAIN. If you set a domain name with both methods, the -dn option takes precedence.
-UserName -un	user_name	Required if the domain uses Native or LDAP authentication. User name to connect to the domain. You can set the user name with the -un option or the environment variable INFA_DEFAULT_DOMAIN_USER. If you set a user name with both methods, the -un option takes precedence. Optional if the domain uses Kerberos authentication. To run the command with single sign-on, do not set the user name. If you set the user name, the command runs without single sign-on.
-Password -pd	password	Required if you specify the user name. Password for the user name. The password is case sensitive. You can set a password with the -pd option or the environment variable INFA_DEFAULT_DOMAIN_PASSWORD. If you set a password with both methods, the password set with the -pd option takes precedence.
-CloudProvisioningConfigurationID -cpcid	cloud_provisioning_configuration_id	Required. ID of the cloud provisioning configuration to update with the certificate file path.
-CertificateFilePath -certPath	certificate_file_path	Required. Path to the ADLS Service Principal certificate on the Data Integration Service machine.
-SecurityDomain -sdn	security_domain	Required if the domain uses LDAP authentication. Optional if the domain uses native authentication or Kerberos authentication. Name of the security domain to which the domain user belongs. You can set a security domain with the -sdn option or the environment variable INFA_DEFAULT_SECURITY_DOMAIN. If you set a security domain name with both methods, the -sdn option takes precedence. The security domain name is case sensitive. If the domain uses native or LDAP authentication, the default is Native. If the domain uses Kerberos authentication, the default is the user realm specified during installation.
-ResilienceTimeout -re	timeout_period_in_seconds	Optional. Amount of time in seconds that infacmd attempts to establish or re-establish a connection to the domain. If you omit this option, infacmd uses the timeout value specified in the INFA_CLIENT_RESILIENCE_TIMEOUT environment variable. If no value is specified in the environment variable, the default of 180 seconds is used.

Author

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